Ngoc-Dung Nguyen

AI / MACHINE LEARNING RESEARCHER

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AI/ML researcher combining **deep-learning engineering** with a **biomedical & cancer-science foundation**. Five years building predictive models on clinical, imaging and prospective-cohort data — peer-reviewed **neural-network survival analysis** for cancer risk, a transformer multi-cancer risk model **deployed as a live clinical decision-support app**, and a **3D multi-parametric MRI pipeline** for liver fibrosis staging under cross-vendor domain shift. Hands-on with PyTorch, Transformers, nnU-Net and deep survival models, plus a strong statistical and medical-domain background. Working across both halves of clinical intelligence: **medical foundation models** (PEFT fine-tuning of SAM-Med3D in a semi-supervised co-training loop) and **LLM/RAG systems** (retrieval-augmented document QA with Hugging Face Transformers under 4-bit quantization). Seeking to apply biomedical AI to **clinical intelligence** — where the evaluation discipline that time-to-event and cross-vendor work demands is what generative clinical systems most lack.

PROFESSIONAL EXPERIENCE

AI Researcher — *MoAdata* **Jan 2022 – Present**

- Develop **deep-learning models** predicting cancer risk from large-scale prospective cohort and clinical imaging data — data pipeline through training, evaluation and deployment.
- **CHORALE — liver fibrosis staging under vendor shift** (MICCAI 2026 CARE-Liver challenge). Built the full 3D multi-parametric MRI pipeline: nnU-Net segmentation, ShaSpec shared-specific fusion for missing modalities, radiomics-GBM ensemble, ordinal staging heads, Optuna tuning. **Held-out unseen-vendor AUC 0.8334** for cirrhosis detection; liver segmentation **Dice 0.9332 in-domain / 0.9409 unseen vendor**. Shipped as GPU Docker containers to the official leaderboard, which scored both splits.
- **CanAttend** — transformer/attention multi-task survival model for 10 cancer types on KCPS-II (development risk) and SEER (post-diagnosis survival); 5/10/15-year horizons, IPCW C-index; best-in-panel discrimination for lung, prostate, liver and kidney cancer. Attention analysis surfaced biomarker interactions (creatinine clearance rate, lipid profile).
- **Deployed CanAttend end-to-end** as an interactive clinical decision-support application — real-time, interpretable risk and survival trajectories (canattend.streamlit.app).
- Built and open-sourced **nDeep**, a deep survival-analysis model for time-to-event risk prediction (github.com/ngocdung03/nDeep).

Public Health / Data Researcher — *National Cancer Center, South Korea* **May 2020 – Dec 2021**

- Applied **survival analysis and machine learning** to prospective epidemiological data to model cancer risk, resulting in peer-reviewed publications.
- Integrated and cleaned large biomedical datasets; built statistical-analysis plans, Cox proportional-hazards models, and reproducible R/Python pipelines.
- Produced conference posters and manuscripts; presented at an international conference (APHA, 2018).

Researcher — *Vietnam Institute of Applied Medicine* **Jun 2016 – Aug 2017**

- Designed research plans and clinical-trial questionnaires; contributed to two research projects and two publications.

Research Assistant — *National COPD Prevention Project, Bach Mai Hospital* **Jan 2016 – Jun 2016**

- Built study questionnaires and collected patient data; led data-quality management — supervised interviews and cleaned raw data.

SELECTED PUBLICATIONS

Manuscript in preparation, 2026. Nguyen ND, et al. "CHORALE: Cross-Contrast Harmonization of Representations, Aggregated for Liver Evaluation in Cross-Vendor Fibrosis Staging." Role: method design, all experiments, evaluation protocol, manuscript.

TECHNICAL SKILLS

Deep Learning
PyTorch, Transformers, CNNs/LSTMs, nnU-Net, SAM-Med3D, deep survival analysis

Medical Imaging
Multi-parametric MRI, 3D segmentation, radiomics, domain generalization / OOD evaluation

LLM & Retrieval
Retrieval-augmented generation, LangChain, Chroma vector stores, semantic chunking, quantized local inference (4-bit NF4)

ML & Statistics
scikit-learn, survival analysis (Cox PH), IPCW C-index, model calibration, ensembling, Optuna, TensorFlow

Programming
Python, R, SQL, STATA, ArcGIS

Tools & Platforms
Git, Docker, AWS, Streamlit, Jupyter, LaTeX

Domain
Cancer epidemiology, liver disease imaging, cohort/clinical data, medical prognosis, biostatistics, clinical NLP

EDUCATION

MSc, Financial Engineering
WorldQuant University, USA
2021–2024 · GPA 91/100

MSc, Public Health
NCC Graduate School of Cancer Science & Policy, South Korea
2017–2019 · GPA 4.12/4.30

BSc, Public Health
Hanoi Medical University, Vietnam
2011–2015 · GPA 7.64/10

LANGUAGES

English **IELTS 7.0**
Korean **KIIP Level 5**

Manuscript in preparation, 2026. **Nguyen ND**, Lee J-H, Bae C-Y. "CanAttend: A Transformer-based Clinical Decision Support System for Multi-Cancer Risk Prediction and Survival Analysis using Attention Mechanisms." Role: model design, training, evaluation, web application, manuscript.

2023 Chulyoung B., Boseon K., Sunha J., Jonghoon L., **Dung NN**. "A Study on Survival Analysis Methods Using **Neural Networks** to Prevent Cancers." *Cancers.* Role: deep survival modeling, data cleaning, analysis, manuscript.

2021 **Dung NN**, Jinhee K., Mikyung K. "Metabolic health and central obesity are significantly associated with an increased risk of thyroid cancer: Data from the Korean Genome and Epidemiology Study." *Cancer Epidemiology, Biomarkers & Prevention.* Role: research questions, data cleaning, survival analysis, manuscript.

2018 Huy NV, Mai DL, Thanh NV, **Dung NN**, Anh NT. "A Systematic Review of Effort–Reward Imbalance among Health Workers." *Int. Journal of Health Planning and Management.*

2017 Huy NV, **Dung NN**, Hanh LT, Thanh NV. "Patient satisfaction with health-care services at a national institute of ophthalmology." *Int. Journal of Health Planning and Management.*

2016 Huong LT, …, **Dung NN**, et al. "Cigarette Smoking among Adolescents aged 13–15 in Viet Nam: Results from GYTS 2014 Data." *Asian Pacific Journal of Cancer Prevention.*

Conference: APHA Annual Meeting & Expo, 2018 — HPV cohort & progression of cervical intraepithelial neoplasia (survival analysis).

SELECTED PROJECTS

CHORALE — Cross-Vendor Liver Fibrosis Staging (MICCAI 2026 CARE-Liver)

- 3D multi-parametric MRI (T1, T2, DWI, GED1–4) fibrosis staging and liver segmentation with missing modalities: nnU-Net v2, ShaSpec fusion, SAM-Med3D, radiomics-GBM ensemble, TotalSegmentator ROI. Leave-one-vendor-out generalization study including documented negative results for domain-adversarial training, style alignment and mutual learning. Dockerized submission; 460 training / 280 test cases.

CanAttend — Transformer Multi-Cancer Risk & Survival CDSS

- Attention-based multi-task survival model on KCPS-II and SEER; baselines DeepSurv, DeepHit and Random Survival Forest; IPCW C-index evaluation with competing events. Live application: canattend.streamlit.app · github.com/ngocdung03/CanAttend

RAG Chatbot — Retrieval-Augmented Document QA *(team project)*

- End-to-end retrieval-augmented generation over PDFs: semantic chunking at embedding-similarity breakpoints, Chroma vector retrieval, conversational memory, and file-versioned prompt templates composed as a LangChain Expression Language pipeline. Served Vicuna-7B under 4-bit NF4 quantization (Hugging Face Transformers, BitsAndBytes), fitting a 7B model onto consumer-class GPUs. github.com/aio25-mix002/rag-chatbot-streamlit

nDeep — Deep Survival Analysis Model

- Deep-learning model for time-to-event cancer-risk prediction on prospective cohort data; neural network for survival analysis. github.com/ngocdung03/nDeep

Machine-Learning Risk-Factor Analysis (NCC)

- Applied ML algorithms in R/Python to identify predictive factors of disease from metabolic and epidemiological data, contributing to a peer-reviewed study.

RESEARCH PROPOSALS

Retinal Imaging and AI for Dementia Risk Prediction — *proposal, 2026*

- *Study design only — no model trained, no results reported.* Specifies a multimodal design for predicting incident dementia (ICD-10-confirmed) from non-mydriatic colour fundus photography in the Canadian Longitudinal Study on Aging Comprehensive Cohort (n ≈ 30,000). Three candidate pipelines compared — intermediate image/metadata fusion, machine-to-machine transfer, and segmentation-guided prediction — with an image-quality screening stage and an evaluation framework separating discrimination from calibration.

Vietnamese	**Native**

RELEVANT CERTIFICATIONS

- AI for Medical Prognosis — *deeplearning.ai*
- AWS Machine Learning Fundamentals — *Udacity*
- Deep Learning with TensorFlow — *Codecademy*
- Docker & Kubernetes
- Advanced Statistical Methods in Python
- Mathematics for Machine Learning — *Imperial College London*
- Predictive Analytics Nanodegree — *Udacity*

PROFESSIONAL DEVELOPMENT

- **AI VIETNAM — AIO2025**, intensive AI/ML program (2025)
- Kaggle Masterclass; SQL & Python data-science tracks
- Good Clinical Practice; Writing in the Sciences (Stanford)

AWARDS

2018 · Overseas Academic Conference Scholarship, National Cancer Center (Korea)

2015 · Active Student of the Whole Course, Hanoi Medical University

2011–12 · Total Scholarship, Hanoi Medical University